

Carnivores

PHYLUM	Chordata
CLASS	Mammalia
ORDER	Carnivora
FAMILIES	16
SPECIES	279

CLASSIFICATION NOTE

As the evolution of the order Carnivora becomes better known, scientists have subdivided existing families, and included those from the order Pinnipedia (seals and walrus). There are currently 16 families, of which 14 are shown in this book.

Dogs and relatives see pp.168–75	Red Panda see p.190
Bears see pp.176–81	Mustelids see pp.191–6
Sea lions see pp.182–3	Malagasy carnivores see p.197
Walrus see p.184	Mongoose see p.198
Seals see pp.184–5	Civets and relatives see p.199
Skunks see p.186	Cats see pp.200–7
Raccoons and relatives see pp.187–8	Hyenas see pp.208–9

Although the term **carnivore** is commonly used to describe an animal that eats meat, it also refers specifically to members of the order Carnivora. While most members of the group eat meat, some have a mixed diet or are entirely herbivorous. Meat-eating carnivores are the dominant predators on land in all habitats: their bodies and lifestyles are highly adapted for hunting. However, there is great variety within the group, which includes species as diverse as the giant panda and the walrus. Uniquely among mammals, carnivores have 4 carnassial teeth. They also have a penis bone (baculum). Indigenous to most parts of the world, carnivores have also been introduced to Australasia.

Hunting

Carnivores include some of nature's most skillful and efficient predators. Most use keen senses of sight, hearing, or smell to locate prey, which they catch either by pouncing from a concealed place or by stalking and then running down their quarry in a lengthy chase or swift rush. Many can kill animals larger than themselves. Weasels kill by biting the back of the head and cracking the skull, while cats bite into the neck, damaging the spinal cord, or into the throat, causing suffocation. Though some carnivores are solitary, others hunt in packs.



SOLITARY HUNTER

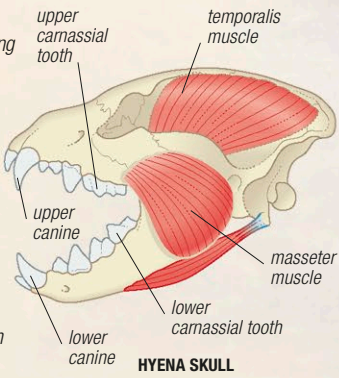
The bobcat, which feeds mainly on small prey such as the snowshoe hare, hunts alone.

Anatomy

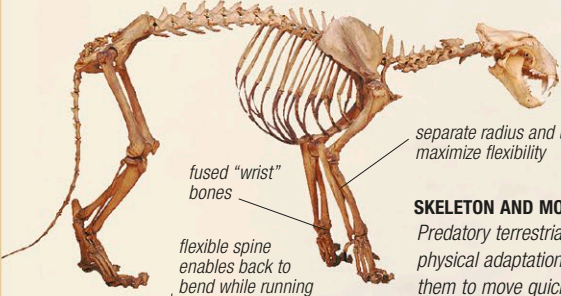
Although carnivores vary considerably in size and shape, most share several features that make them well suited to a hunting lifestyle. A typical terrestrial carnivore is a fast and agile runner with sharp teeth and claws, acute hearing and eyesight, and a well-developed sense of smell. Carnassial teeth (see below) are present in predacious living carnivores but are less well developed among omnivorous, herbivorous, and some piscivorous species. Carnivores have either 4 or 5 digits on each limb. Members of the cat family have sharp, retractable claws used to rake prey, defend themselves, and climb. Most other carnivores have nonretractable claws, often used for digging.

JAWS AND TEETH

Most carnivores have sharp teeth and powerful jaws for killing and disemboweling prey. The temporalis muscles, which are most effective when the jaws are open, are used to deliver a powerful stab from the sharp canines. The carnassial teeth are sharpened molars in the upper and lower jaws. They act like scissors and slice through hide and flesh, and are used to crush bone, too. In combination with the masseter muscles, which can be used when the jaw is almost completely closed, they form a powerful shearing tool for tearing flesh.



HYENA SKULL



TIGER SKELETON

SKELETON AND MOVEMENT

Predatory terrestrial carnivores have physical adaptations that enable them to move quickly over the ground in pursuit of prey. The spine is generally flexible, the limbs are relatively long, and the collarbone is reduced, maximizing the mobility of the shoulders. To help increase the length of their stride and stability, all carnivores have fused wrist bones, and dogs and cats walk on their toes (rather than the soles of the feet).



SIBERIAN TIGER

HUNTING IN PACKS

Lions generally hunt in groups to capture large animals. The females (males rarely join in) usually stalk to within 98 ft (30 m) and encircle the prey. After a short charge, the animal is brought down with a grab to the flank, then killed by suffocation with a bite to the throat.

Social groups

Although many carnivores live alone or in pairs, others form groups that take different forms, and have complex structures. Lion prides, for example, consist of several related families, although most males leave the pride into which they are born. Lions spend most of their time together, hunt cooperatively, and tend each other's young. In most other groups, individual ties are looser. Red and Arctic foxes live in groups of one adult male and several vixens, but each adult hunts alone in a different part of the group's territory. Elephant seals gather in large numbers only during breeding. The groups are made up of several males and their respective harems of females, which are closely guarded.



PLAY

Young carnivores develop their fighting skills through play. By playing together, these red foxes learn to test another animal's strength without suffering painful consequences.

SHARED PARENTHOOD

These young slender-tailed meerkats are not necessarily the offspring of the adult watching over them. Sharing parental duties is common in many carnivore societies.